REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Graphing Data Practice Questions

Grade 9 Advanced | Module 1 Lesson 4 | Graphing Data

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: Slope = $\Delta y/\Delta x$; linear: $y = mx + b$; quadratic: $y = kx^2$; inverse: $y = k/x$.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 17–22.

P01. Which variable is deliberately changed by the investigator?

A. Independent variable B. Dependent variable
C. Controlled variable D. Random error

P02. The dependent variable is usually plotted on the:

A. Horizontal axis B. Vertical axis
C. Page border D. Graph title only

P03. A straight line through the origin is evidence of:

A. An inverse relationship B. A constant nonzero intercept
C. Direct proportionality D. A quadratic relationship

P04. For y proportional to x^2, doubling x makes y:

A. Twice as large B. Half as large
C. Unchanged D. Four times as large

P05. A best-fit line should:

A. Represent the overall trend in the data B. Join every point by sharp segments
C. Always pass through the origin D. Avoid all measured points

---

P06. A straight graph of force F (N) against extension x (cm) passes through (1, 2) and (5, 14). Calculate its slope.

A. 3 N/cm B. 6 N/cm
C. 1.5 N/cm D. -3 N/cm

P07. A measured relationship is $y = 4x + 3$, where x is in seconds and y is in metres. Find y when x = 4 s.

A. 38 m B. 19 m
C. 9.5 m D. -19 m

P08. A data model is $y = 2x^2$, with x in seconds and y in metres. Calculate y at x = 3 s.

A. 36 m B. 9 m
C. 18 m D. -18 m

P09. A data model is y = 48/x. Find y when x = 6. Both x and y are dimensionless.

A. 16 B. 4
C. -8 D. 8

P10. Three repeated measurements are 12, 14 and 16 s. Calculate their mean.

A. 14 s B. 28 s
C. 7 s D. -14 s

P11. Define a controlled variable.

P12. What does the slope of a graph represent?

P13. What does a nonzero y-intercept mean?

P14. How can a graph indicate an inverse relationship?

P15. Why should axis labels include units?

P16. A straight graph of force F (N) against extension x (cm) passes through (1, 2) and (7, 20). Calculate its slope.

P17. A straight graph of force F (N) against extension x (cm) passes through (1, 2) and (9, 26). Calculate its slope.

P18. A measured relationship is y = 6x + 3, where x is in seconds and y is in metres. Find y when x = 4 s.

P19. A measured relationship is y = 8x + 3, where x is in seconds and y is in metres. Find y when x = 4 s.

P20. A data model is $y = 2x^2$, with x in seconds and y in metres. Calculate y at x = 4 s.

P21. A data model is $y = 2x^2$, with x in seconds and y in metres. Calculate y at x = 5 s.

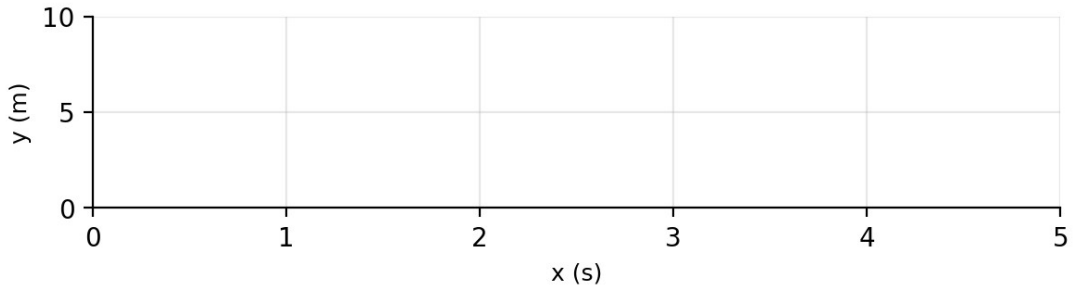
P22. A data model is y = 72/x. Find y when x = 9. Both x and y are dimensionless.

P23. A data model is y = 96/x. Find y when x = 12. Both x and y are dimensionless.

P24. Three repeated measurements are 13, 15 and 17 s. Calculate their mean.

P25. Three repeated measurements are 14, 16 and 18 s. Calculate their mean.

P26. Plot x = 0, 1, 2, 3, 4 s against y = 1, 3, 5, 7, 9 m. Draw a best-fit line and state its slope and intercept.



P27. A student calls the data x = 1, 2, 3 and y = 3, 6, 9 an inverse relationship. Evaluate the claim.

P28. Describe a graphing procedure for noisy experimental data.

P29. Data follow $y = 4x^2$. Suggest a transformation that makes a straight-line graph.

P30. An apparent straight line does not pass through the origin although theory predicts proportionality. Suggest two checks.